



MMDG-DTI: Drug–target interaction prediction via multimodal feature fusion and domain generalization

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ABSTRACT

Recently, deep learning has become the essential methodology for Drug–Target Interaction (DTI) prediction. However, the existing learning-based representation methods embed the prior knowledge encapsulated by the training data and, as such, acquire redundant domain information irrelevant to the DTI prediction, compromising the generalization capability of a deep network to unfamiliar instances. To tackle this limitation, we propose a novel DTI prediction method using Multi-Modal feature fusion and Domain Generalization (MMDG-DTI). Specifically, we first use pre-trained Large Language Models (LLMs) to obtain generalized textual features covering nearly the whole biological text vocabulary, with the capacity to handle unseen samples and extract robust and discriminative features. In parallel, we propose a unified structural feature extractor based on a hybrid Graph Neural Network (GNN). The extractor improves the performance of MMDG-DTI by extracting features from another modality and discarding the representation of domain-specific prior substructures. Then, we integrate the complementary information of LLM and GNN features using an innovative classifier, while enhancing the generalization ability with the help of Domain Adversarial Training (DAT) and contrastive learning. Last, we propose a novel cross-domain evaluation protocol to enhance the predication fidelity in practical applications. The quantitative and visualization results obtained on several benchmarking datasets demonstrate that MMDG-DTI achieves state-of-the-art performance under both single-domain and cross-domain settings, confirming the superiority of the proposed method. The datasets and codes will be available on the project homepage: <https://github.com/JU-HuaY/MMDG-DTI>.

1. Introduction

Drug–Target Interaction (DTI) prediction is critical to virtual drug screening and improving the efficiency of traditional drug discovery pipelines [1]. The existing DTI prediction models can be roughly divided into two categories: classification-based [2] and regression-based methods [3]. A classification model identifies the drug and target protein interactions, while a regression model quantifies the binding affinity. Since both categories have similar backbones, we cross-refer to the existing DTI and Drug–Target binding Affinity (DTA) prediction models in the literature [4,5]. Similar to other tasks in the pattern recognition field, most of the existing DTI methods are deep-learning-based [6,7]. These methods have explored a variety of frameworks by investigating the intrinsic properties of drug compounds and target proteins and integrating diverse deep neural network components. Öztürk et al. first proposed DeepDTA [7], which applies Convolutional Neural Networks (CNN) and Multi-Layer Perception (MLP) to extract drug

and protein features for DTA estimation. Along these lines, numerous studies [8–10] have focused on enhancing DTI models by refining three critical components: drug feature extraction, protein feature extraction, and classifier design.

For drug feature extraction, Lee et al. [8] proposed to use molecular fingerprints [11] and MLP, which achieves a better prediction performance than using the sequential drug features in DeepDTA. Nguyen et al. [9] proposed a graph structure to represent drug features and investigated a variety of graph networks, including GCN [12], GAT [13], and GIN [14], further enriching the process of drug feature extraction. For protein feature extraction, Tsubakiet al. proposed CPI-GNN [10] to represent proteins via word embedding that contains more complex textual information than the amino acid embedding used in DeepDTA, improving the prediction performance to some extent. To enhance the textual property of proteins, Chen et al. [15] proposed

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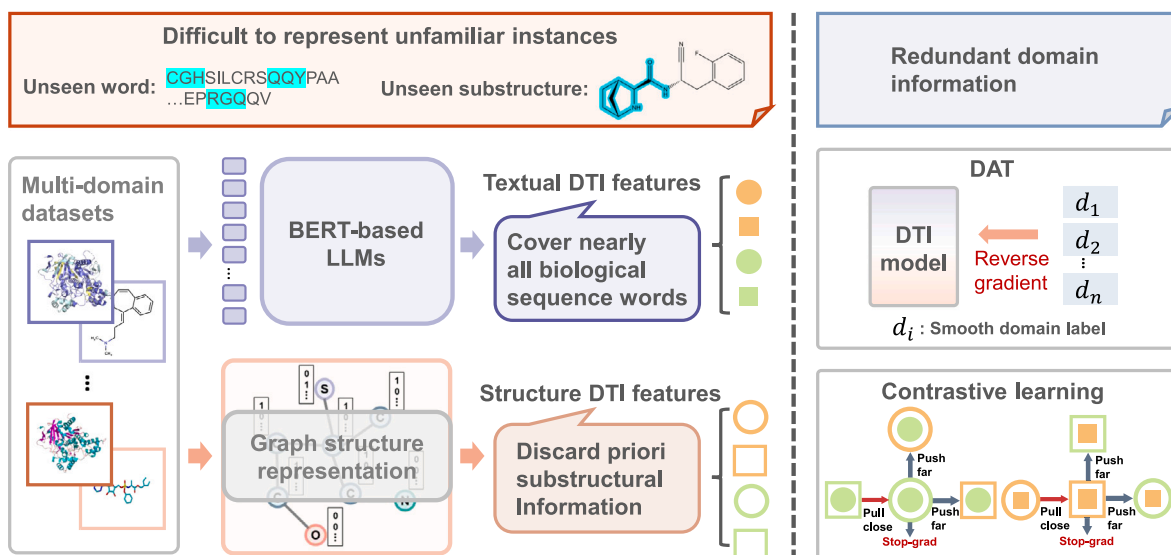


Fig. 1. The left part presents the solutions to the current limitations of the learnt representations, and the right part displays how to mitigate the negative effects of domain redundant information. In contrastive learning, shapes represent data domains and colors represent DTI labels.

TransformerCPI, which uses Word2Vec [16] for protein feature representation and employs Transformer [17] for final feature extraction. This work improves the DTI prediction performance on single-domain datasets. In addition, many studies have improved the DTI prediction through classifier design. Both HADTI [18] and MATT-DTI [19] apply attention layers to strengthen the interactive information among the drug and protein features. MCANet [20] developed multi-head cross attention to further enhance the attention module. DeepCDA [21] uses an LSTM [22] module prior to the attention layer to capture the temporal information of samples, further complementing the beneficial information for the classification layers.

The above methods have achieved promising results despite using lightweight model architectures. Later, the continual development of hardware and GPU resources further promotes the DTI research area. Recent methods usually use multi-scale or multimodal features for high-performance DTI prediction. MGraphDTA [23], built upon GraphDTA, enhances the feature extraction by combining multi-scale protein and drug features. MolTrans [24] uses a knowledge-inspired sub-structural pattern mining algorithm to decompose structural features on a finer scale for more accurate and interpretable DTI prediction. CPIInformer [1] employs two modalities, molecular fingerprints and graph structures, to represent drug features and extract DTI information via Informer [25], uplifting the performance of DTI prediction. DeepGS [26] is another example that employs two modalities for drug feature representation. The method proposed to represent the drug molecules using Smi2Vec and graph structures demonstrated an effective prediction performance. Furthermore, based on CPIInformer, MFR-DTA [27] incorporates amino acid embedding for protein feature extraction. All the above methods verify that the use of multimodal features is beneficial for DTI prediction.

Although the above DTI prediction methods have achieved satisfactory results on single-domain datasets, they may have limited generalization ability when dealing with unseen samples. As shown in Fig. 1, the inadequacy is mainly attributed to the domain properties of the training data and the representation capacity of the trained model. Specifically, existing word embeddings and structure representations, such as Word2Vec [15], Smi2Vec [26] and GNN-based models [9, 28], incorporate prior knowledge about the samples restricted to the source domain training dataset. While these methods can demonstrate significant predictive capabilities on a single-domain dataset, the representation process reduces the generalization to unfamiliar instances containing unseen words or substructures, leading to a performance

degradation. On the other hand, the existing benchmarking datasets have distinctive domain properties, such as sequence recordings, sample feature distributions, and statistical situations. Specifically, the records of a sample are not exactly the same in different databases due to discrepancies in data integrity standards, variations in collection criteria, and inconsistencies in processing software. These properties could mislead a model to acquire redundant knowledge that is irrelevant to DTI prediction during the training process, thereby reducing its generalization capacity.

To address these issues, we propose a novel DTI prediction model driven by multimodal feature fusion and domain generalization (MMDG-DTI). **First**, we use two BERT-based Large Language Models (LLMs), *i.e.*, Prot-Bert (Prot_Bert_bfd) [29] and Smiles-Bert (PubChem10M_SMILES_BPE_450K) [30], for protein and drug feature extraction, respectively. The two models are trained on nearly four hundred billion amino acid sequences and ten million chemical molecules, respectively. Due to the use of huge training data, the textual features derived from the two LLMs cover nearly the whole biological text vocabulary and exhibit better generalization capabilities, compared to the previous methods. We also apply DenseNet [31] and Informer [25] to refine the learned textual features for further performance boosting. **Second**, we propose a hybrid Graph Neural Network (GNN) to improve the DTI prediction performance by extracting structural features from another modality. The extracted features reflect the structural relationship between atoms in drugs and amino acids in proteins, and perform excellently on small sample datasets. Besides, the structural feature extractor uses a uniform amino acid embedding and molecular graph structures without a domain-specific prior sub-structural knowledge. Thus, this extractor can process samples with unseen substructures. **Third**, we design an innovative classifier to fuse the semantic features of LLM-based and GNN-based features for more comprehensive DTI prediction. The classifier mitigates the negative impact of redundant domain information on the model by integrating two domain generalization methods: Domain Adversarial Training (DAT) [32] and contrastive learning [33]. Besides, we propose novel cross-domain experimental protocols to better evaluate the performance of DTI prediction methods in terms of the generalization capability.

We evaluate the proposed method under both single-domain and cross-domain settings. The experimental results demonstrate a superior DTI prediction accuracy of the proposed method over the existing state-of-the-art approaches, especially under the cross-domain setting. We also examine the effectiveness of each innovative component proposed

in our MMGD-DTI by conducting various ablation experiments. Last, we demonstrate the strong generalization capability of the proposed method by visualizing the high-response regions of protein–drug pairs. Extensive experimental results show that our proposed method is effective and can provide a better option for drug discovery researchers, but it has room for improvement in cross-domain experiments. Meanwhile, the designed cross-domain experiments provide a more realistic environment for evaluating drug–target interaction models.

In summary, the main contributions of MMDG-DTI include:

- A large-language-model-based textual feature extractor for protein and drug feature representations with enhanced generalization capability.
- A hybrid-GNN-based structural feature extractor that supplements topological features.
- An innovative classifier that fuses the textual and structural features, while mitigating the overfitting issue by domain generalization.
- A design of novel cross-domain experiments for the evaluation of DTI prediction methods in terms of generalization capability.

The rest of this paper is organized as follows. We first present the related work in Section 2. Then, we introduce the proposed MMDG-DTI method in Section 3.2. Last, we report and analyze the experimental results in Section 4 and draw the conclusion in Section 5.

2. Related work

In this section, we present the prior work related to the proposed method, with a focus on DTI prediction and domain generalization.

2.1. DTI prediction

Drug–target interaction prediction from sequential data has evolved through three main phases: traditional machine learning [34], deep learning based on single features [20] and deep learning based on multimodal features [27]. At the beginning, researchers used hand-crafted methods to represent the drug and target features, and employed shallow machine learning methods such as KNN [35] and SVM [36] to predict DTIs. However, the performance of these methods is relatively limited, both in terms of accuracy and generalization capability. With the development of deep learning, a number of advanced DTI prediction methods such as the DeepDTA [7], DeepConvDTI [8], HADTI [18], MATT-DTI [37], and MCANet [20] have been proposed.

While the above studies use single representations to extract drug and target features, recent studies have found that multimodal feature fusion is more effective [27,38,39]. Specifically, the raw data formats of both protein and drug are biologically meaningful sequences, inherently containing two salient properties: biological and textual. Regarding the proteins, not only do individual amino acids possess distinct biological properties, but different amino acid groupings also form structural domains with specialized significance. Based on this biological property, MFR-DTA [27] proposed a BioCNN block that extracts both global and individual features to optimize protein features. The textual properties of proteins are experimentally driven. For instance, CPI-GNN [10] adopts three amino acids as a word, while TransformerCPI [15] captures this feature via Word2Vec [16]. Despite the textual property lacking natural language interpretability, it positively impacts on the predictive performance of deep models. Similarly, the drug properties contain different modalities such as biological properties in the form of molecular fingerprints, textual meanings in the form of SMILES and the graph structure, which describes direct atomic positional relationships within the molecule. Numerous existing approaches [27,38] improve the DTI prediction by integrating drug features from multiple modalities. However, the quality of the features varies across modalities, and feature refinement methods tend to affect the generalization capability of the trained model, which merits further research and exploration.

2.2. Domain generalization methods

Domain generalization [40] was first introduced by Blanchard et al. as a machine learning problem that addresses the out-of-distribution (OOD) scenarios where target data is inaccessible during model training [41]. The main domain generalization methods include: domain alignment [42], meta-learning [43], data augmentation [44], self-supervised learning [45], and regularization strategies [46]. These domain generalization methods have typically been studied in two settings: multi-source domain generalization with multiple distinct but relevant domains and single-source domain generalization with homogeneous training data drawn from a single domain [41]. Most of the existing studies focus on the multi-source setting, which assumes multiple distinct but relevant domains are available. As presented in [40], the original motivation for studying domain generalization is to leverage multi-source data to learn representations invariant to different marginal distributions. Multiple domains enable the discovery of stable cross-domain patterns that could transfer better. In contrast, single-source domain generalization relates to OOD robustness [47], examining model robustness under perturbations. Single-source methods do not require domain labels and could apply to multi-source scenarios. In fact, most deep-learning-based models are generic OOD solutions with both single and multi-source experiments [48].

We use two domain generalization methods, Domain Adversarial Training (DAT) [32] and contrastive learning [33], to improve the generalization capability of the proposed method. DAT is a multi-source domain generalization method that requires domain labels, whereas contrastive learning is a single-source domain generalization approach that does not necessitate domain labels. Considering the properties of the current benchmarking datasets, we design cross-domain experiments with two different protocols corresponding to multi-source and single-source domain generalization scenarios. Therefore, we use only the contrastive learning strategy in the single-source protocol and a combination of both domain generalization methods in the multi-source protocol.

3. Materials and methods

3.1. The benchmarking datasets

We evaluate the proposed method on six public datasets, including four classical DTI benchmarks, Human (H) [1], C.elegans (C) [1], BindingDB (B) [1], DrugBank (D) [18], and two DTA-based datasets, Davis (Da) [18] and KIBA (K) [18]. For the four classical DTI benchmarks, predicted parameters are the interaction labels, which indicate the interactive relationship between a pair of samples. These labels are obtained from the corresponding database. The predicted parameters in two DTA-based datasets are constructed based on drug target affinity according to thresholds. Specifically, Zhao et al. [18] constructed the two binary classification datasets by setting the thresholds 5.0 and 12.1 for the Davis and KIBA datasets, respectively. For single-domain tasks, our evaluation approach and dataset division follows previous research works [1,18]. Human and C.elegans datasets are evaluated by the 5-fold cross-validation protocol, and the BindingDB is divided into training, validation, and test sets. Consistent with [18], we also perform 10 repeated 5-fold cross-validation to evaluate the performance of a model on DrugBank, Davis, and KIBA. To ensure that the result is statistically significant, we use 5 random seeds to train the model and report the mean and variance of the scored results in all experimental tables. The statistics of the six benchmarking datasets are summarized in Table 1.

For cross-domain tasks, we design two types of experiments, multi-source and single-source cross-domain experiments based on the six datasets mentioned above. We design multi-source cross-domain experiments based on four classical DTI benchmarks, using three of them as the source domains and the other as a target domain. With this design, the multi-source cross-domain experiments consist of four protocols,

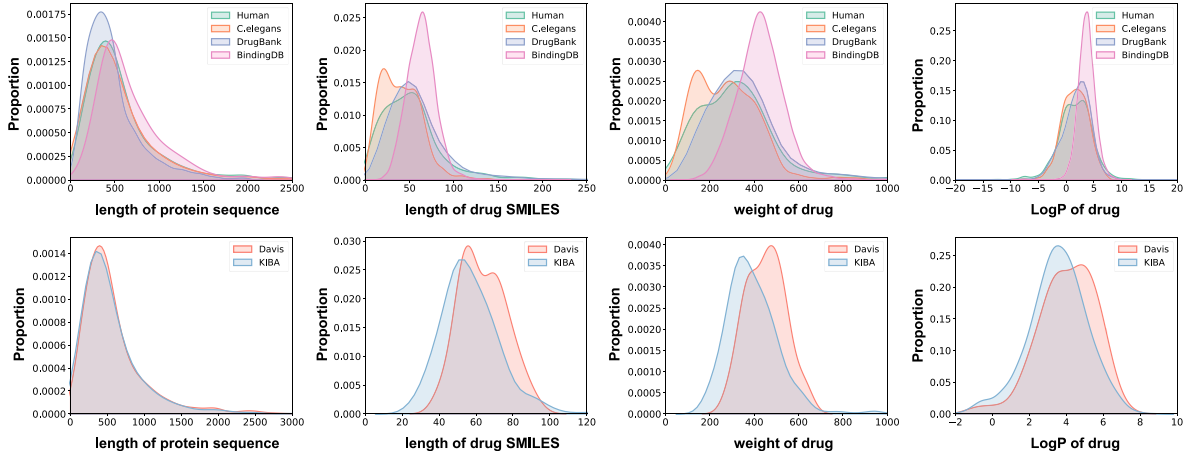


Fig. 2. The protein length, atom numbers of the drugs, and the distribution of the molecular properties, including weights, and the oil–water partition coefficient (LogP) in the six datasets.

Table 1
Statistics of the benchmarking datasets.

Datasets	Protein	Drug	Interaction	Positive	Negative
Human	852	1052	6212	3369	2843
C.elegans	2504	1434	7511	4000	3511
BindingDB	811	49567	60780	33777	27003
DrugBank	4254	6645	35022	17511	17511
Davis	379	68	25772	7320	18452
KIBA	225	2068	116350	22154	94196

namely B&D&H to C, B&D&C to H, B&H&C to D, and D&H&C to B. Due to the Davis and KIBA datasets share similar label properties, we use these two datasets to design single-source cross-domain experiments, including two protocols, namely Da to K and K to Da. In addition, we present the basic information of the benchmarks in Fig. 2, including the length of the protein sequence and drug SMILES, the weight of the drug, and the drug oil–water partition coefficient: LogP. Both Table 1 and Fig. 2 show that there are differences not only in the number of proteins and drugs, but also in the distribution of drug properties across these datasets, especially in the four classical DTI benchmarks. Despite the relatively similar distribution of data between the Davis and KIBA datasets, the large variation in the number of drugs still makes the cross-domain experiments challenging.

3.2. The pipeline of MMDG-DTI method

The pipeline of our MMDG-DTI method is shown in Fig. 3. It has five main components: textual feature extractor, structural feature extractor, classifier, and two domain generalization modules. In the rest of this section, we introduce these three components in more detail.

3.2.1. The LLM-based textual feature extractor

As shown in Fig. 3, we design the textual feature extractor based on two BERT-based language models, a DenseNet encoder, and the Informer block. First, we use Prot-Bert [29] and Smiles-Bert [30] to represent protein sequences and drug SMILES as embedding features, $\mathbf{F}_p \in \mathbb{R}^{L_p \times C_p^p}$ and $\mathbf{F}_d \in \mathbb{R}^{L_d \times C_d^d}$, respectively. Both Prot-Bert and Smiles-Bert are BERT-based Large Language Models (LLMs) trained on huge sequence data using self-supervised learning, with the benefit that their embeddings have strong generalization ability. Then, we extract the textual features of the protein and drug embeddings by the DenseNet-based encoder. The encoder is similar to the one proposed in our previous work, CPInformer [1], which can capture both the global and local features of the sequence very well. The encoder is formulated as:

$$\mathbf{F}^{i+1} = \mathbf{F}^i + g * \text{Relu}(\text{Bn}(\text{Conv}_{1D}(\mathbf{F}^i))), \quad (1)$$

where $\mathbf{F}^i \in \mathbb{R}^{l \times c}$ is the feature map obtained by the i th layer, l and c are the length and representation channel of the embedding features, respectively. g is the retention rate, $\text{Relu}()$ is the ReLU activation function, $\text{Bn}()$ is the batch normalization layer, and $\text{Conv}_{1D}()$ is the one-dimensional convolution layer with a kernel size of 15. Last, we extract the textual DTI features, $\mathbf{F}_t \in \mathbb{R}^{L_t \times C_h}$, via the Informer block:

$$\mathbf{F}_t = I(\mathbf{F}_p, \text{Concat}(\mathbf{F}_D, \mathbf{F}_P)) + I(\mathbf{F}_D, \text{Concat}(\mathbf{F}_D, \mathbf{F}_P)), \quad (2)$$

$$I(\mathbf{Q}, \mathbf{F}_C) = \text{Softmax}\left(\frac{\tilde{\mathbf{Q}}\mathbf{F}_C^T}{\sqrt{d}}\right)\mathbf{F}_C, \quad (3)$$

where $\mathbf{F}_p \in \mathbb{R}^{L_p \times C_h}$ and $\mathbf{F}_D \in \mathbb{R}^{L_d \times C_h}$ are the protein and drug textual features, respectively. All the dimension information of different features is listed in Table 2. $\text{Concat}()$ and $I()$ are the concatenate operation and the Informer function. $\mathbf{F}_C \in \mathbb{R}^{L_t \times C_h}$ is the concatenated DTI feature. Regarding the Informer module, we follow the approach used in CPInformer. Specifically, Informer replaces the query matrix $\mathbf{Q}$ in the Transformer module [17] with a sparse matrix $\tilde{\mathbf{Q}}$. For more details of Informer, a reader can refer to [1,25].

3.2.2. The hybrid-GNN-based structural feature extractor

To complement the DTI model with the structural relationships of drugs and target proteins, we design the structural feature extractor based on hybrid Graph Neural Networks (GNN). The extractor represents a protein via the amino acid embedding [27] and then extracts the protein feature, $\mathbf{F}_p \in \mathbb{R}^{L_p \times C_s^p}$, via the DenseNet-based encoder. Similarly, the extractor represents a drug by atom features, $\mathbf{F}_d \in \mathbb{R}^{L_d \times C_s^d}$, and the adjacency matrix, $\mathbf{A}_d \in \mathbb{R}^{L_d \times L_d}$, based on the graph structure of the drug via RDKit [49].

We then extract the structural DTI features based on the above three feature matrices. Note that we can form a drug molecule into a topological graph with atoms as nodes through the graph representation. However, the target protein sequences cannot directly form a graph structural relationship between amino acids. Although we can use deep-learning-based protein structure prediction methods to accomplish this work, it was accompanied by consuming excessive computational resources. More importantly, a small prediction error may have a large negative impact on the performance of the DTI model. Therefore, we construct the protein feature matrix with amino acids as nodes, while the relationship design between nodes is to record the adjacent amino acids as neighboring nodes based on the sequence order. By design, we can use the one-dimensional Deep-Wise Convolution (DWC) [50] with the convolution kernel of 3 to extract the structural relationships of the proteins. We construct the association matrix, $\mathbf{C} \in \mathbb{R}^{L_p \times L_d}$, between a pair of samples by multiplying the relevant protein and atom features.

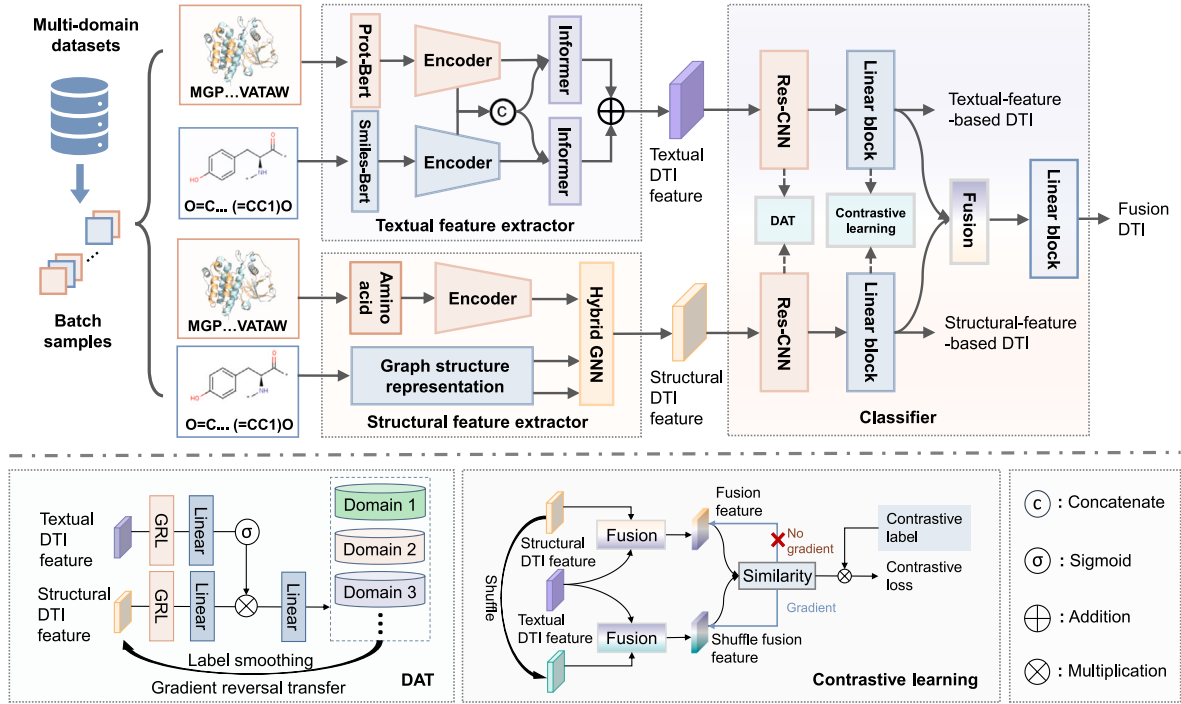


Fig. 3. The overall pipeline of MMDG-DTI. It has five main components: textual feature extractor, structural feature extractor, classifier, and two domain generalization modules, namely Domain Adversarial Training (DAT) and contrastive learning.

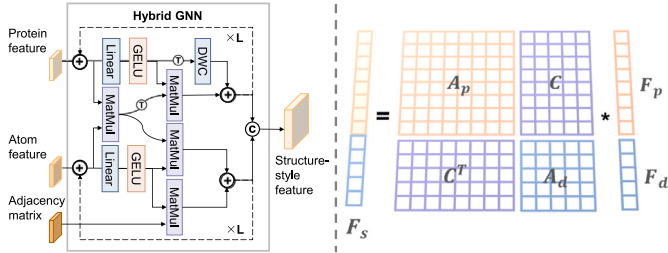


Fig. 4. The architecture of the proposed hybrid GNN module (left) and the formulation of our hybrid GNN (right).

This matrix and its transpose, $C^T \in \mathbb{R}^{L_d \times L_p}$, are then used as the weights for atom and protein features, respectively.

As shown in Fig. 4, we can extract the structural DTI features, $F_s \in \mathbb{R}^{L_s \times C_h}$, via the hybrid GNN, which is formulated as follows:

$$F_s = \begin{bmatrix} A_p & C \\ C^T & A_d \end{bmatrix} \begin{bmatrix} F_p \\ F_d \end{bmatrix}, \quad (4)$$

$$A_p F_p = DW C(F_p), C = \text{Softmax}(F_p F_d^T), \quad (5)$$

where A_p is the assumed amino acid adjacency matrix. Specifically, the structural DTI feature is the result after N hybrid GNN aggregations. During each aggregation, F_p and F_d are updated as follows:

$$\delta(F_p) = DW C(\omega(F_p)) + C \omega(F_d) + F_p, \quad (6)$$

$$\delta(F_d) = C^T \omega(F_p) + A_d \omega(F_d) + F_d, \quad (7)$$

$$F_s = \text{Concate}(\delta(F_p), \delta(F_d)), \quad (8)$$

where $\delta()$ is the update operation, and $\omega()$ is the combination of a linear layer and the GELU function. For smoother feature extraction, we add residual operations to update F_p and F_d . In fact, the hybrid GNN captures the relationships between drugs and target proteins by

integrating the strengths of several different components. These include the extraction of structure relationships of drugs using a classical GNN, proteins using a DWConv, and interactions using association matrices. As mentioned above, the fusion of the three types of structure relationships is infinitely approximated to the graph structure features of protein-drug interactions based on larger adjacency matrices. However, the proposed hybrid GNN is more efficient and refined.

3.2.3. The innovative classifier

In this part, we integrate textual and structural DTI features to predict DTI. As shown in Fig. 3, the classifier consists of five basic blocks: Res-CNN, Linear, Fusion, DAT, and contrastive learning. First, we use Res-CNN blocks to further refine textual and structural DTI features. The Res-CNN block includes one-dimensional convolution, ReLU, residual connection, and global average pooling layers. Then, the two refined features are used to predict the textual-feature DTI and structural-feature DTI via the Linear block. In the next step, we fuse the two semantic features, $f_t \in \mathbb{R}^{C_h}$ and $f_s \in \mathbb{R}^{C_h}$, which are obtained from the last layer of Linear blocks. Further, a new Linear block is used to predict the fused DTI, based on the fused features, $f \in \mathbb{R}^{C_h}$. The fusion process is formulated as:

$$f = \omega(\text{Concate}(f_t, f_s, \sigma(f_s) * f_t, \sigma(f_t) * f_s)), \quad (9)$$

where $\omega()$ and $\text{Concate}()$ are the linear layer and concatenate operation. $\sigma()$ is the Sigmoid function. Last, we aggregate the three predicted DTIs as the final prediction result.

For cross-domain experiments, we employ two domain generalization modules: Domain Adversarial Training (DAT) [32] fusion and contrastive learning [33] to further improve the generalization of the model. As shown in Fig. 3, the DAT fusion acts on the refined features, $f_{rc} \in \mathbb{R}^{C_h}$, generated by the Res-CNN and is trained by the smooth labels of the data domain. In forward propagation, the DAT acts as an identity transform. In backpropagation, the DAT takes the gradient of the next layer, multiplies it by a negative hyperparameter, and passes it to the previous layer. By design, the negative effects of any data domain differences can be mitigated in the refined features.

The inference based on the textual feature, $f_{trc} \in \mathbb{R}^{C_h}$, and structural feature, $f_{src} \in \mathbb{R}^{C_h}$, can be formulated as:

$$\mathbf{y}_p = \omega(\sigma(\omega_l(d(f_{trc}))) * \omega_s(d(f_{src}))), \quad (10)$$

where $\mathbf{y}_p \in \mathbb{R}^3$ is the inferred result, $\omega()$ and $d()$ are the linear layer and DAT operation, and $\sigma()$ is the Sigmoid function. For more details of DAT, readers can refer to [32].

It is noteworthy that textual DTI features extracted from LLMs have similar distributions and greater generalization across datasets [29]. In contrast, structural DTI features, which are determined by molecular structures, weights, and chemical properties, vary much more across dataset domains [51]. Thus, structural DTI features may lead to fusion features that are negatively influenced by redundant domain-specific information. We use contrastive learning [33] to suppress these negative influences. The process is shown in Fig. 3. We first randomly shuffle the order of structural DTI features in a batch and fuse it with the relevant textual DTI features as their shuffle fusion features, $f^{i*} = \text{fuse}(\mathbf{F}_{ts}^i, \mathbf{F}_{ss*}^i) \in \mathbb{R}^{C_h}$, according to Eq. (9). Next, we calculate the similarity of the shuffle fusion features and the fusion features, $f^i = \text{fuse}(\mathbf{F}_{ts}^i, \mathbf{F}_{ss}^i) \in \mathbb{R}^{C_h}$, and multiply the similarity with the contrast label content as the contrast loss, L_{contra} , described as below:

$$L_{contra} = \sum_{B}^{i=1} Eq(\mathbf{x}^i, \mathbf{x}^{i*}) \cdot \text{Sim}(\text{stopgrad}(f^i), f^{i*}) \quad (11)$$

$$Eq(\mathbf{x}^i, \mathbf{x}^{i*}) = \begin{cases} +1, \text{label}(\mathbf{x}^i) == \text{label}(\mathbf{x}^{i*}), \\ -1, \text{otherwise}, \end{cases} \quad (12)$$

where $Eq(\mathbf{x}^i, \mathbf{x}^{i*})$ is the contrast label, $\mathbf{F}_{ts} \in \mathbb{R}^{B \times C_h}$, $\mathbf{F}_{ss} \in \mathbb{R}^{B \times C_h}$ and $\mathbf{F}_{ss*} \in \mathbb{R}^{B \times C_h}$ are a batch of textual semantic features, structural semantic features and shuffled structural semantic features. $\mathbf{x}^i$ and $\mathbf{x}^{i*}$ are the i th index of the normal and shuffled batch, respectively. B is the batch size, $\text{Sim}(a, b) = -\frac{a}{\|a\|} \cdot \frac{b}{\|b\|}$ is cosine similarity, $\text{stopgrad}()$ is stop-gradient operation. $\text{label}(\mathbf{x}^i)$ is the i th DTI label. By design, the shuffle fusion features, f^* , are guided to go close to or far away from their corresponding anchors, f , based on the DTI label information as shown in Fig. 1. As a result, the DTI semantic weight of the fusion feature can be increased, while the sensitivity to domain information is effectively reduced.

3.3. Loss functions and implementation details

The proposed MMDG-DTI method is a multi-task learning model that uses different loss functions to train different tasks performed by dedicated branches. We use the Binary cross-entropy loss function to calculate the three DTI losses, including the textual-feature and structural-feature DTI losses, L_B^t and L_B^s , and the fusion DTI loss, L_B^f , according to:

$$L_B = \frac{1}{N} \sum_i -[y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)], \quad (13)$$

where y_i denotes the label of sample i , the positive class is 1 and the negative class is 0. p_i denotes the probability that sample i is predicted to be a positive class. For the DAT fusion block training, we use the Cross-entropy Loss function:

$$L_C = -\frac{1}{N} \sum_i \sum_{c=1}^M y_{ic} \log(p_{ic}), \quad (14)$$

where p_{ic} is the predicted probability that sample i belongs to category c , y_{ic} indicates whether the category of sample i is c , and M and N are the total number of classes and samples respectively. For contrastive learning, the loss, L_{contra} , is calculated by Eq. (11). Last, we obtain the final loss as:

$$L_{total} = \lambda_1 * L_B^t + \lambda_2 * L_B^s + \lambda_3 * L_B^f + \lambda_4 * L_C^{domain} + \lambda_5 * L_{contra}, \quad (15)$$

where each λ is a balancing parameter.

Table 2

Hyper-parameter settings of MMDG-DTI.

Hyper-parameter	Value
The representation channels $C_t^p, C_t^d, C_s^p, C_s^d$	1024, 768, 75, 75
The hidden-layer feature channel C_h	75
The feature length L_p, L_d, L_s, L_{ss}	1200, 100, 1300, 1300
The retention rate $G_t^p, G_t^d, G_s^p, G_s^d$	128, 128, 15
The encoder kernel size $K_t^p, K_t^d, K_s^p, K_s^d$	15, 15, 31
The Res-CNN kernel size K	7
Number of convolution layers in Res-CNN blocks	3
Number of linear layers in Res-Linear blocks	3
Single-domain task: $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$	1, 1, 1, 0, 0
Single-domain task (Unseen samples): $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$	1, 1, 1, 0, 1
Single-source cross-domain task: $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$	1, 1, 1, 0, 1
Multi-source cross-domain task: $\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$	1, 1, 1, 1, 1

We use the Area Under Receiver Operating Characteristic Curve (AUC-ROC) and Area Under Precision-Recall Curve (AUC-PR) as our main evaluation metrics, consistent with the previous work [1]. Specifically, the ROC curve plots the true positive rate along the Y-axis and false positive rate on the X-axis. The PR curve plots the precision and recall values on Y-axis and X-axis, respectively. Besides, we use the precision and recall metrics for single-domain experiments by following standard evaluation protocols. Our method was implemented in Python 3.8 with PyTorch 1.8.0. The experiments were carried out on a machine with the Ubuntu 20.04 operating system, Intel Core i7-11700K CPU and one NVIDIA GeForce RTX 3090 card. We used the AdamW optimizer [52] for network training. In the training process, we set the learning rate as 5×10^{-4} , the weight decay as 7×10^{-2} , the batch size, as 16, and the dropout ratio as 0.1. More hyperparameter settings are listed in Table 2.

4. Experimental results and discussion

In this section, we first present the effectiveness of each proposed innovative component by the results of an ablation study. Then, we compare the proposed method with the state-of-the-art methods on both single-domain and cross-domain tasks. Last, we display a scatter plot of different extracted semantic features and visualize the high-response regions of protein-drug pairs for an interpretable verification of our method.

4.1. Ablation study

For the ablation study, we evaluate the effectiveness of each innovative component on both the single-domain and cross-domain tasks. In particular, for the single-domain tasks, we evaluate the effectiveness of the main components, including the Textual-Feature Extractor (TFE), the Structural-Feature Extractor (SFE), and the fusion module in the classifier. For the cross-domain tasks, we also evaluate the effectiveness of the proposed two domain generalization modules, Domain Adversarial Training (DAT) and Contrastive Learning (CL). It should be noted that the cross-domain tasks are divided into two types: multi-source and single-source.

4.1.1. Ablation study on the single-domain task

The MMDG-DTI model is a newly designed framework without a corresponding baseline method. Therefore, we use the performance of the textual-feature DTI and the structural-feature DTI on four classical DTI benchmarks to evaluate the corresponding extractors. Note that unrelated modules are not involved in the training and inference when evaluating a single extractor. In addition, we also compare different semantic feature fusion approaches to prove the effectiveness of the proposed fusion module. The compared semantic feature fusion approaches mainly include concatenation, pairwise inner product [53], cross product [54], element-wise product [55], and the Informer-based

Table 3

The ablation study results (AUC-ROC) obtained on four benchmarks, namely Human, C.elegans, BindingDB and DrugBank.

Model	TFE	SFE	Fusion Approach	Human	C.elegans	BindingDB	DrugBank
Model-1	✓		–	0.979 ± 0.003	0.983 ± 0.002	0.956 ± 0.003	0.879 ± 0.004
Model-2		✓	–	0.980 ± 0.003	0.985 ± 0.004	0.951 ± 0.005	0.876 ± 0.005
Model-3	✓	✓	Concatenation	0.983 ± 0.002	0.988 ± 0.003	0.963 ± 0.003	0.886 ± 0.004
Model-4	✓	✓	Pairwise inner product	0.976 ± 0.005	0.983 ± 0.005	0.953 ± 0.004	0.878 ± 0.006
Model-5	✓	✓	Cross product	0.973 ± 0.007	0.979 ± 0.007	0.943 ± 0.006	0.880 ± 0.005
Model-6	✓	✓	Element-wise product	0.984 ± 0.002	0.989 ± 0.002	0.959 ± 0.002	0.889 ± 0.003
Model-7	✓	✓	Informer-based fusion module	0.988 ± 0.003	0.991 ± 0.003	0.966 ± 0.001	0.894 ± 0.002
Model-8	✓	✓	The proposed fusion module	0.991 ± 0.002	0.996 ± 0.002	0.971 ± 0.001	0.901 ± 0.001

Table 4

The ablation study results (AUC-ROC) obtained using the multi-source cross-domain protocols, B&D&H to C, B&D&C to H, D&H&C to B, and B&H&C to D.

Model	TFE	SFE	DAT	CL	B&D&H to C	B&D&C to H	D&H&C to B	B&H&C to D
Model-mc1	✓			–	0.893 ± 0.003	0.763 ± 0.002	0.583 ± 0.003	0.539 ± 0.002
Model-mc2	✓		✓	–	0.907 ± 0.002	0.781 ± 0.003	0.617 ± 0.001	0.551 ± 0.003
Model-mc3		✓		–	0.877 ± 0.005	0.741 ± 0.004	0.525 ± 0.006	0.479 ± 0.005
Model-mc4		✓	✓	–	0.895 ± 0.004	0.735 ± 0.003	0.527 ± 0.002	0.493 ± 0.003
Model-mc5	✓	✓			0.917 ± 0.003	0.790 ± 0.002	0.652 ± 0.002	0.575 ± 0.004
Model-mc6	✓	✓	✓		†0.919 ± 0.002	†0.789 ± 0.001	†0.649 ± 0.003	†0.580 ± 0.002
Model-mc7	✓	✓		✓	†0.921 ± 0.002	†0.794 ± 0.003	†0.657 ± 0.001	†0.579 ± 0.002
Model-mc8	✓	✓	✓	✓	†0.927 ± 0.002	†0.799 ± 0.001	†0.658 ± 0.002	†0.583 ± 0.003

fusion module [1]. Among them, the first four approaches are basic operations between two vectors, and the Informer-based fusion module is the semantic feature fusion approach based on Informer [1]. The experimental results are shown in Table 3.

Model-1 and Model-2 represent the models using only the textual and structural DTI features, respectively. It is clear that Model-1 performs better on the larger sample size datasets, BindingDB and DrugBank, while Model-2 is superior on small sample sizes. Since the representation of the textual feature extractor is pre-trained on a large amount of data, the extracted features appear to have absorbed more information and exhibit stronger generalization. Thus, the extractor is more powerful when there are more different proteins and drug molecules in the dataset. Although the structural features contain less information, they can highlight molecular structural styles, which may promote slightly better performance on small datasets.

We use the remaining models, from Model-3 to Model-8, to present the performance of different fusion approaches. Concatenation is a simple and effective method to combine textual and structural features. The accuracy of Model-3 is higher than that of the models (Model-1 and Model-2) using single features. The performance of both Model-4 and Model-5 is relatively poor, indicating that both the pairwise inner product and cross product are less suitable for semantic feature fusion. The performance of Model-6 and Model-3 is similar, confirming that the element-wise product retains the content of the two fused semantic features well. The performance of Model-7 shows that the Informer-based fusion module [1,25] significantly improves the DTI model by introducing attention information. Most importantly, the proposed fusion module fuses the semantic features with fewer parameters, and achieves the state-of-the-art on the four benchmarks.

4.1.2. Ablation study of the cross-domain task

For cross-domain task, we first present the results of the multi-source cross-domain protocols in Table 4. The first four models in the table are single DTI feature models. It can be seen that the generalization of the textual-feature-based DTI is much better than that of the structural-feature-based DTI, with the results obtained by Model-mc1 and Model-mc3. This result is consistent with the expectation, since the textual feature extractor received more samples for pre-training. Besides, we can see from the performance of both Model-mc2 and Model-mc4 that DAT helps to optimize the model in cross-domain experiments noticeably. Since contrastive learning is designed using the fusion of textual and structural features, it is only applied to the two-stream model. The performance of Model-mc5 shows that the

Table 5

The ablation study results (AUC-ROC) obtained using single-source cross-domain protocols, including Da to K and K to Da.

Model	TFE	SFE	CL	Da to K	K to Da
Model-sc1	✓			0.625 ± 0.006	0.685 ± 0.007
Model-sc2		✓		0.587 ± 0.010	0.607 ± 0.012
Model-sc3	✓	✓		0.638 ± 0.008	0.708 ± 0.010
Model-sc4	✓	✓	✓	†0.644 ± 0.007	†0.716 ± 0.008

use of the two-feature fusion improves the model generalization. In most cases, incorporating DAT or contrastive learning improves the generalization of the model, except for Model-mc6 on the protocol D&H&C to B. To understand the reasons, we analyze Fig. 2 and find that the distribution of samples in the BindingDB dataset is quite different from the other three datasets. Moreover, the domain properties of the other three datasets are similar, making it difficult for DAT to be effective. However, when we use DAT and contrastive learning together, the gradient transformation brought by contrastive learning alleviates this problem to some extent, making MMDG-DTI optimal for the multi-source cross-domain task.

Next, we present the results of the single-source cross-domain protocols in Table 5. The main difference between the two datasets, KIBA and Davis, is the number of unique drug molecules included. Table 1 clearly shows that the number of unique drug molecules in KIBA is much larger than that of Davis, thus the DTI models perform better on the K to Da protocol than that on the Da to K protocol. In addition, Table 5 again demonstrates that textual DTI features generalize better than structural DTI features. When the two are combined, the overall performance could be further improved. This means that the information content is complementary. Moreover, contrastive learning can improve the model in the single-source cross-domain task, as shown by the performance of Model-mc5.

4.2. Comparison with the state-of-the-art methods

In this part, we compare the proposed MMDG-DTI method with the state-of-the-art DTI prediction models, including DeepConvDTI [8], TransformerCPI [15], MolTrans [24], HADTI [18], CPInformer [1], MCANet [20], TriMulDTI [56], and MFR-DTA [27], on the four classical DTI benchmarks and the two DTA-based threshold-segmented datasets. The comparative experiment covers both single-domain and cross-domain tasks.

Table 6

A comparison with the state-of-the-art methods in terms of AUC-ROC, AUC-PR, precision, and recall, on the Human, C.elegans, BindingDB and DrugBank datasets.

Method	Human				C.elegans			
	AUC-ROC	AUC-PR	Precision	Recall	AUC-ROC	AUC-PR	Precision	Recall
DCDTI [8]	0.970 ± 0.003	0.978 ± 0.003	0.942 ± 0.002	0.953 ± 0.003	0.979 ± 0.003	0.981 ± 0.002	0.947 ± 0.002	0.950 ± 0.001
TransCPI [15]	0.973 ± 0.002	0.979 ± 0.002	0.916 ± 0.006	0.925 ± 0.006	0.988 ± 0.002	0.987 ± 0.001	0.957 ± 0.006	0.953 ± 0.005
MolTrans [24]	0.975 ± 0.004	0.977 ± 0.003	0.952 ± 0.003	0.935 ± 0.003	0.987 ± 0.003	0.989 ± 0.002	0.961 ± 0.005	0.967 ± 0.003
HADTI [18]	0.978 ± 0.002	0.980 ± 0.003	0.932 ± 0.002	0.938 ± 0.003	0.985 ± 0.003	0.987 ± 0.003	0.961 ± 0.004	0.967 ± 0.003
CPInformer [1]	0.985 ± 0.003	0.989 ± 0.002	0.961 ± 0.002	0.951 ± 0.002	0.991 ± 0.002	0.992 ± 0.001	0.956 ± 0.002	0.976 ± 0.001
MCANet [20]	0.983 ± 0.003	0.989 ± 0.002	0.961 ± 0.002	0.937 ± 0.002	0.989 ± 0.002	0.990 ± 0.002	0.967 ± 0.002	0.963 ± 0.001
TriMulDTI [56]	0.985 ± 0.003	0.986 ± 0.002	0.953 ± 0.001	0.949 ± 0.001	0.986 ± 0.003	0.988 ± 0.002	0.957 ± 0.002	0.961 ± 0.001
MFR-DTA [27]	0.986 ± 0.003	0.983 ± 0.002	0.938 ± 0.001	0.959 ± 0.001	0.988 ± 0.002	0.985 ± 0.001	0.968 ± 0.002	0.969 ± 0.002
MMDG-DTI	0.991 ± 0.002	0.993 ± 0.002	0.978 ± 0.002	0.945 ± 0.002	0.996 ± 0.001	0.996 ± 0.001	0.979 ± 0.001	0.981 ± 0.001

Method	BindingDB				DrugBank			
	AUC-ROC	AUC-PR	Precision	Recall	AUC-ROC	AUC-PR	Precision	Recall
DCDTI [8]	0.944 ± 0.002	0.947 ± 0.003	0.933 ± 0.003	0.941 ± 0.002	0.867 ± 0.001	0.777 ± 0.002	0.750 ± 0.001,	0.698 ± 0.001
TransCPI [15]	0.951 ± 0.003	0.949 ± 0.004	0.959 ± 0.003	0.937 ± 0.003	0.837 ± 0.003	0.836 ± 0.002	0.750 ± 0.003	0.792 ± 0.003
MolTrans [24]	0.956 ± 0.002	0.949 ± 0.003	0.921 ± 0.002	0.918 ± 0.003	0.861 ± 0.002	0.836 ± 0.002	0.786 ± 0.002	0.796 ± 0.002
HADTI [18]	0.956 ± 0.003	0.951 ± 0.004	0.919 ± 0.003	0.931 ± 0.003	0.889 ± 0.001	0.897 ± 0.001	0.829 ± 0.001	0.799 ± 0.001
CPInformer [1]	0.965 ± 0.002	0.966 ± 0.003	0.957 ± 0.002	0.970 ± 0.005	0.887 ± 0.002	0.891 ± 0.002	0.816 ± 0.002	0.823 ± 0.001
MCANet [20]	0.961 ± 0.003	0.963 ± 0.002	0.963 ± 0.003	0.919 ± 0.003	0.897 ± 0.002	0.902 ± 0.005	0.824 ± 0.002	0.827 ± 0.003
TriMulDTI [56]	0.963 ± 0.003	0.967 ± 0.002	0.958 ± 0.004	0.951 ± 0.005	0.895 ± 0.002	0.897 ± 0.003	0.827 ± 0.002	0.819 ± 0.002
MFR-DTA [27]	0.962 ± 0.002	0.959 ± 0.003	0.947 ± 0.003	0.936 ± 0.003	0.893 ± 0.002	0.891 ± 0.003	0.831 ± 0.002	0.806 ± 0.002
MMDG-DTI	0.971 ± 0.001	0.972 ± 0.002	0.960 ± 0.003	0.955 ± 0.002	0.901 ± 0.001	0.897 ± 0.001	0.798 ± 0.002	0.840 ± 0.001

Table 7

A comparison with state-of-the-art methods for the unseen drug, target, and pair settings in the BindingDB dataset. The results designated by ‘*’ are obtained by methods using the MMDG-DTI with the domain generalization modules.

Method	Unseen drug		Unseen target		Unseen pair	
	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR
DCDTI [8]	0.875 ± 0.003	0.843 ± 0.003	0.871 ± 0.004	0.847 ± 0.005	0.851 ± 0.010	0.831 ± 0.013
TransCPI [15]	0.870 ± 0.005	0.824 ± 0.006	0.868 ± 0.005	0.845 ± 0.005	0.831 ± 0.015	0.818 ± 0.017
MolTrans [24]	0.869 ± 0.008	0.845 ± 0.010	0.877 ± 0.011	0.853 ± 0.008	0.831 ± 0.019	0.809 ± 0.020
HADTI [18]	0.870 ± 0.007	0.843 ± 0.006	0.875 ± 0.009	0.850 ± 0.012	0.842 ± 0.013	0.823 ± 0.012
CPInformer [1]	0.881 ± 0.005	0.874 ± 0.004	0.883 ± 0.010	0.862 ± 0.008	0.845 ± 0.015	0.827 ± 0.016
MCANet [20]	0.905 ± 0.003	0.901 ± 0.004	0.892 ± 0.008	0.869 ± 0.009	0.867 ± 0.012	0.838 ± 0.013
TriMulDTI [56]	0.895 ± 0.007	0.898 ± 0.008	0.897 ± 0.009	0.879 ± 0.011	0.862 ± 0.015	0.825 ± 0.0177
MFR-DTA [27]	0.898 ± 0.005	0.899 ± 0.003	0.897 ± 0.006	0.873 ± 0.008	0.863 ± 0.010	0.832 ± 0.011
MMDG-DTI	0.914 ± 0.004	0.910 ± 0.003	0.921 ± 0.006	0.904 ± 0.007	0.872 ± 0.009	0.837 ± 0.012
MMDG-DTI*	0.928 ± 0.003	0.925 ± 0.002	0.926 ± 0.005	0.910 ± 0.006	0.883 ± 0.008	0.847 ± 0.010

4.2.1. A comparison on the single-domain task

For the single-domain task, we use AUC-ROC, AUC-PR, precision and recall as the evaluation metrics. The experimental results on four classical DTI benchmarks are presented in Table 6. According to the table, the proposed MMDG-DTI method achieves superior performance over all the other methods in most metrics. For the Human and C.elegans, which are relatively small datasets, MMDG-DTI outperforms all the other approaches in terms of AUC-ROC, AUC-PR and precision, confirming its superiority in small datasets. Although the recall metrics on the Human dataset are not optimal, this could be mitigated in practical applications by optimizing the thresholds. For the BindingDB dataset, which is divided into training, validation and test sets, we evaluate MMDG-DTI in the same way as in [1], so the metrics on this dataset are presented in Table 6 without variance. The proposed method achieves the best result in terms of AUC-ROC and AUC-PR, with a performance gain of 0.6% compared to the second-best method, CPInformer. The two methods, MCANet and CPInformer, perform better on the two metrics, precision and recall, respectively. However, our method is relatively more balanced on these two metrics. In addition, we report the performance of each model in this dataset for the unseen drug, target, and pair settings in Table 7. Overall, the unseen sample pair settings are more difficult to process than the unseen drug or target settings alone. However, the results also show that the proposed method has advantages in open scenarios because the combination of multiple modal representations makes the model more robust, especially when domain generalization modules are used simultaneously. Furthermore, our model does not perform as well on

the DrugBank dataset as it does on the other datasets, but it achieves the best performance in the AUC-ROC and Recall metrics, demonstrating its competitiveness.

Next, we evaluate the proposed method in terms of AUC-ROC and AUC-PR, on the DTA-based Davis and KIBA datasets. The results are presented in Fig. 5. In the figure, DC-DTI and TransDTI stand for DeepConvDTI [8] and TransformerCPI [15]. Fig. 5 provides a more intuitive performance of these methods, including accuracy and stability, in the box plot format. We can find that other methods often struggle to offer both advantages simultaneously, except for the proposed MMDG-DTI method. In addition, according to Table 1 and Fig. 5, it is clear that Davis and KIBA have far fewer positive class samples than negative class samples, resulting in all models performing worse on the AUC-PR metric than on the AUC-ROC metric. Nevertheless, several experiments obtained on the two DTA-based threshold-segmented datasets show that MMDG-DTI outperforms the existing methods in all the metrics, further confirming the superiority of the proposed method.

4.2.2. Comparative experiment on the cross-domain task

For cross-domain tasks, we present the experimental result of both multi-source cross-domain and single-source cross-domain tasks in Table 8. We evaluate the proposed method on the cross-domain task in terms of AUC-ROC and AUC-PR, and provide the experimental results for two versions of MMDG-DTI for a fair comparison. Table 8 clearly shows the superiority of our method, achieving the best performance in six cross-domain protocols with both multiple and single sources. It is worth noting that the models with more general representation

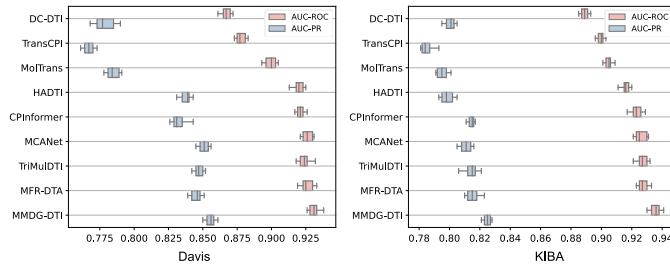


Fig. 5. A comparison with the state-of-the-art methods in terms of AUC-ROC and AUC-PR, on the **Davis** and **KIBA** datasets.

Table 8

A comparison with the state-of-the-art methods in terms of AUC-ROC, AUC-PR, in the cross-domain experiments. The results designated by ‘**’ are obtained by methods using the MMDG-DTI with the domain generalization modules.

Method	B&D&H <i>to</i> C		B&D&C <i>to</i> H		D&H&C <i>to</i> B	
	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR
DCDTI [8]	0.891 ± 0.002	0.895 ± 0.003	0.762 ± 0.003	0.764 ± 0.002	0.593 ± 0.003	0.601 ± 0.005
TransCPI [15]	0.843 ± 0.004	0.859 ± 0.006	0.715 ± 0.004	0.721 ± 0.006	0.507 ± 0.005	0.529 ± 0.006
MolTrans [24]	0.865 ± 0.005	0.877 ± 0.006	0.722 ± 0.007	0.739 ± 0.005	0.527 ± 0.007	0.549 ± 0.008
HADTI [18]	0.878 ± 0.004	0.879 ± 0.003	0.741 ± 0.004	0.748 ± 0.003	0.535 ± 0.006	0.557 ± 0.005
CPInformer [1]	0.875 ± 0.005	0.881 ± 0.007	0.742 ± 0.003	0.753 ± 0.003	0.539 ± 0.004	0.542 ± 0.003
MCANet [20]	0.883 ± 0.004	0.899 ± 0.002	0.761 ± 0.002	0.757 ± 0.003	0.609 ± 0.003	0.623 ± 0.002
TriMulDTI [56]	0.881 ± 0.004	0.893 ± 0.006	0.753 ± 0.003	0.767 ± 0.003	0.602 ± 0.003	0.635 ± 0.005
MFR-DTA [27]	0.887 ± 0.003	0.893 ± 0.002	0.751 ± 0.002	0.769 ± 0.001	0.581 ± 0.002	0.605 ± 0.003
MMDG-DTI	0.917 ± 0.003	0.926 ± 0.003	0.790 ± 0.001	0.811 ± 0.002	0.652 ± 0.003	0.693 ± 0.003
MMDG-DTI*	0.927 ± 0.002	0.943 ± 0.003	0.799 ± 0.001	0.827 ± 0.002	0.658 ± 0.002	0.701 ± 0.003
Method	B&H&C <i>to</i> D		Da <i>to</i> K		K <i>to</i> Da	
	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR
DCDTI [8]	0.543 ± 0.003	0.547 ± 0.004	0.607 ± 0.010	0.408 ± 0.012	0.607 ± 0.012	0.525 ± 0.016
TransCPI [15]	0.497 ± 0.005	0.479 ± 0.003	0.525 ± 0.014	0.321 ± 0.017	0.577 ± 0.013	0.516 ± 0.015
MolTrans [24]	0.509 ± 0.006	0.514 ± 0.007	0.516 ± 0.019	0.303 ± 0.016	0.581 ± 0.021	0.496 ± 0.023
HADTI [18]	0.536 ± 0.004	0.541 ± 0.006	0.565 ± 0.015	0.373 ± 0.013	0.659 ± 0.018	0.531 ± 0.020
CPInformer [1]	0.525 ± 0.003	0.516 ± 0.005	0.583 ± 0.012	0.305 ± 0.014	0.601 ± 0.016	0.510 ± 0.015
MCANet [20]	0.541 ± 0.004	0.543 ± 0.002	0.603 ± 0.010	0.393 ± 0.008	0.672 ± 0.013	0.535 ± 0.012
TriMulDTI [56]	0.537 ± 0.003	0.537 ± 0.004	0.601 ± 0.011	0.405 ± 0.012	0.659 ± 0.013	0.529 ± 0.015
MFR-DTA [27]	0.532 ± 0.002	0.525 ± 0.003	0.597 ± 0.010	0.418 ± 0.007	0.613 ± 0.012	0.527 ± 0.015
MMDG-DTI	0.575 ± 0.003	0.561 ± 0.003	0.638 ± 0.008	0.425 ± 0.009	0.708 ± 0.012	0.543 ± 0.013
MMDG-DTI*	0.583 ± 0.003	0.567 ± 0.002	0.644 ± 0.006	0.311 ± 0.008	0.716 ± 0.010	0.521 ± 0.012

methods perform better in cross-domain experiments. For example, DeepConvDTI, which uses only amino acid sequences to represent proteins and Morgan fingerprints [11,57] to represent drug molecules, can achieve very good performance. In contrast, the word representation pre-trained within a single dataset may improve the performance of models such as Transformer and CPInformer in single-domain experiments, but it under-performs in cross-domain settings. In MMDG-DTI, the textual feature extractor uses a universal word representation model that has been pre-trained on a large number of biomolecule samples, covering almost all biological sequence word representations. It should be noted, that the amino acid embedding and graph structure representation in the structural feature extractor discard the substructure representation of the samples, which is applicable to all unseen proteins and drug molecules. Therefore, the combination of the above two extractors enables MMDG-DTI to handle any unseen instances and achieve state-of-the-art performance in cross-domain experiments. In addition, the two domain generalization methods can further improve the performance of the proposed model in cross-domain settings.

4.3. Discussion on the interpretability of MMDG-DTI

4.3.1. Visualization of different features

In this part, we aim to provide interpretable verification of our method by visualization. First, we present Fig. 6, which shows the scatter plots of different semantic features on the source and target domain test sets. These semantic features are dimensionally reduced by PCA to form position coordinates in planar space, which can visualize

the spatial distribution of the sample features extracted by different models. Thus, Fig. 6 can directly show the contribution of the different innovations. Note that we use the protocols, B&D&H *to* C and Da *to* K, which are the first protocols in the multi-source and single-source cross-domain tasks, to produce the scatter plots. The top and bottom parts of Fig. 6 show the results of the two protocols.

In the multi-source cross-domain tasks, textual DTI features are generally more effective than structural DTI features in predicting DTI for test samples. While textual DTI features perform slightly worse to structural DTI features in the source domain, they show a superior level of generalization in the target domain. In fact, structural DTI features efficiently represent the shallow biological structural information of protein and drug molecules. Although the sample pairs can be efficiently represented under single-domain conditions, the oversensitivity to the biological structural information also reduces its performance in cross-domain experiments. In contrast, the textual DTI feature is a materialization of the deep semantic information of the sample pairs, which represents the deep semantic associations between the sample pairs more abstractly and improves its generalization. As expected, the model in predicting DTI is improved by fusing the two features, and the interaction-sample points cluster more closely. Besides, the contrastive learning module significantly improves the feature representation capability, as presented in the visualization. The positive and negative sample points have more distinct class regions in the two principal component dimensions of the extracted features. Such improvement has primarily been achieved by using contrastive learning which helps to extract semantic information from the labels thus leading to more pronounced differences in the features from different classes. Furthermore,

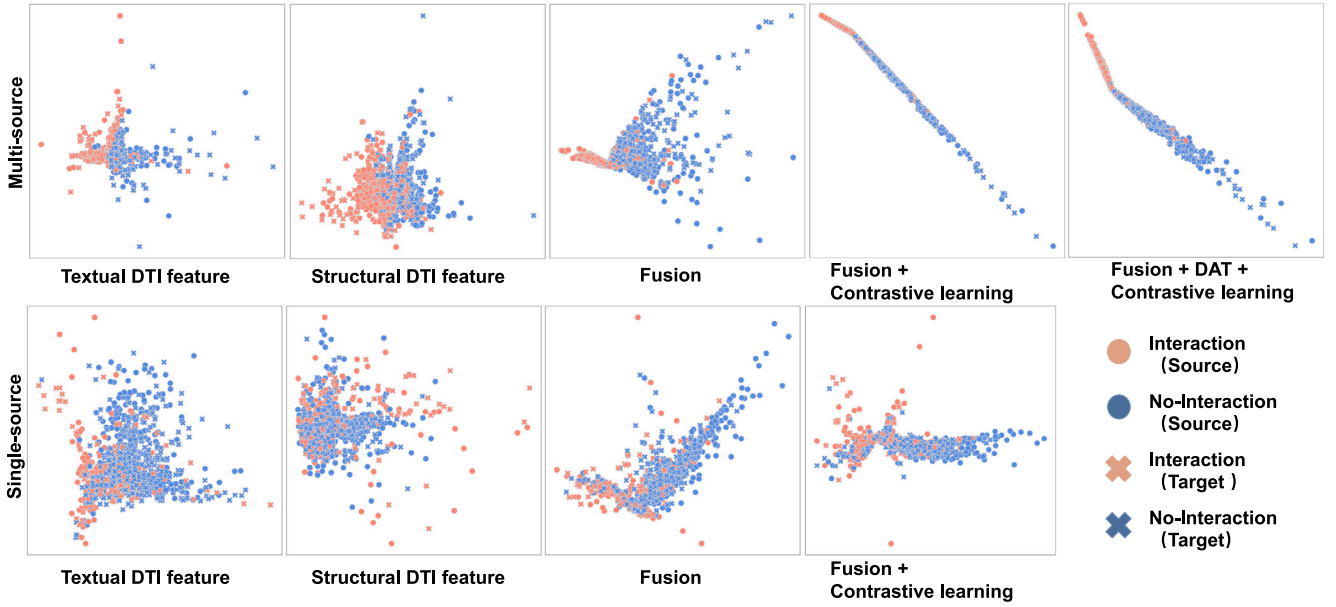


Fig. 6. The contribution of different innovations to cross-domain tasks. The X and Y coordinates for each point in the figure represent the values of each DTI feature after dimensionality reduction to a 2D space through principal component analysis (PCA).

DAT reduces the contamination of the domain information and helps to balance the distribution space between positive and negative sample scatters.

For the single-source cross-domain task, the impact of each innovation on the model is similar to its effect on the multi-source cross-domain task. However, the significant differences in the number of drugs between the Davis and KIBA datasets make it challenging for the model to differentiate between positive and negative samples in the target domain. Although contrastive learning alleviates this problem to some extent, it is still a challenge to design a model that performs well in this cross-domain protocol.

4.3.2. The visualization of the high-response regions

To analyze the perceptual capabilities of the proposed model regarding sample structure, we visualize the high-response regions of the protein–drug pairs in the test sets. As shown in Fig. 7, the visualization includes Source-Domain Sample Visualizations (SDSVs) and Target-Domain Sample Visualizations (TDSVs). SDSVs represent the visualizations of the model responses to source-domain test samples taken from the same dataset as the training data. In contrast, TDSVs depict the responses to the target-domain test samples in cross-domain tasks. Note that the response vectors for both proteins and drugs were derived by computing the mean of the attention weight matrix, $\text{Softmax}(\frac{QF}{\sqrt{d}})$, from the textual feature extractor and the association matrix, C , from the structural feature extractor. Thus, these response vectors contain both the textual semantics and the structural information of the samples. The region with the top 30% response values on each response vector is recorded as the high-response region. Besides, we produce visualizations under identical experimental conditions for the CPInformer [1] to compare with the proposed model. As previously demonstrated [27], these high-response regions may not accurately reflect real-world drug–target binding regions, but they are critical for deep learning models to determine whether there is an interaction between a pair of samples. Therefore, these regions can be used to assess the robustness and generalization of the deep model trained on different datasets. According to Fig. 7, it is clear that the MMDG-DTI models trained on different datasets are more similar in the high-response region when dealing with the same proteins or drugs than that of CPInformer. This indicates that MMDG-DTI captures intrinsically

Table 9

The average similarity of the high-response regions of source-domain and target-domain samples.

Datasets	CPInformer		MMDG-DTI	
	Protein	Drug	Protein	Drug
Human	0.391 ± 0.231	0.573 ± 0.275	0.575 ± 0.263	0.655 ± 0.308
C.elegans	0.463 ± 0.257	0.612 ± 0.318	0.625 ± 0.279	0.713 ± 0.257
BindingDB	0.372 ± 0.214	0.317 ± 0.236	0.553 ± 0.187	0.624 ± 0.271
DrugBank	0.307 ± 0.198	0.373 ± 0.213	0.465 ± 0.190	0.538 ± 0.239
Davis	0.160 ± 0.103	0.232 ± 0.117	0.391 ± 0.213	0.420 ± 0.220
KIBA	0.242 ± 0.125	0.115 ± 0.087	0.453 ± 0.317	0.318 ± 0.253

invariant features that increase its robustness and performance on previously unseen data.

To verify the statistical significance of the above findings, we calculate the response similarity between SDSVs and TDSVs in all the benchmarks and report the results in Table 9. We use the Jaccard similarity coefficient to calculate the similarity of the response vectors and determine the mean and standard deviation based on all the similarity results in the individual benchmarks. The results show that the MMDG-DTI models learned using different training data have higher similarity in response to proteins and drugs on the six benchmark datasets, as compared to CPInformer, confirming the claimed findings.

5. Conclusion

We presented a novel MMDG-DTI model for DTI prediction with better generalization capability across domains. We first constructed a LLMs-based textual feature extractor capable of handling practically all sample instances, to improve the generalization of the model. Then, the structural feature extractor based on a hybrid GNN was proposed to enhance the model by capturing the structural relationships of sample components. Last, we designed an innovative classifier that combined the advantages of textual and structural DTI features and improved generalization with the help of innovative domain generalization modules. To optimize the evaluation, we proposed new cross-domain experiments, including single-source and multi-source cross-domain protocols, based on the existing public datasets. To provide an intuitive understanding of the proposed method, we also included several visualization results.

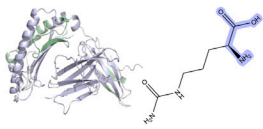
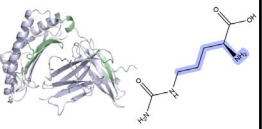
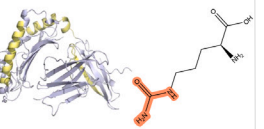
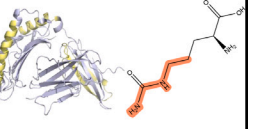
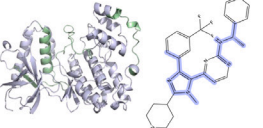
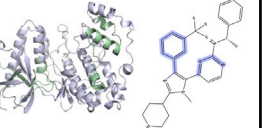
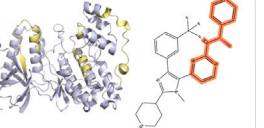
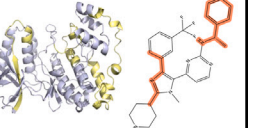
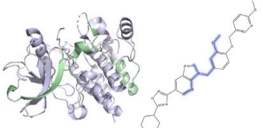
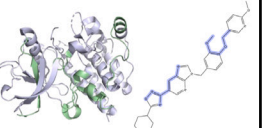
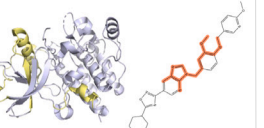
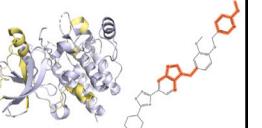
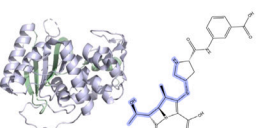
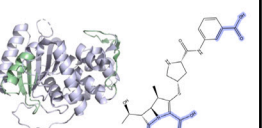
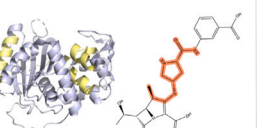
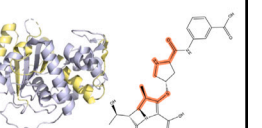
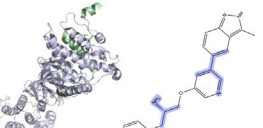
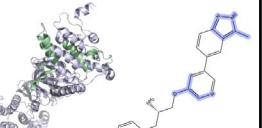
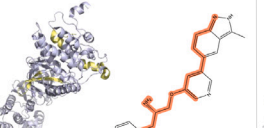
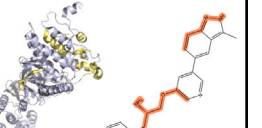
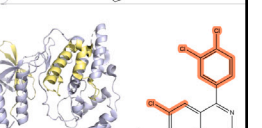
Dataset	The SDSVs of CPInformer	The TDSVs of CPInformer	The SDSVs of MMDG-DTI	The TDSVs of MMDG-DTI
Human P (P01889) D (9750)				
C.elegan P (Q9XXC0) D (447721)				
BindingDB P (P07333) D (118009799)				
DrugBank P (P45059) D (150610)				
Davis P (Q2M2I8) D (11314340)				
KIBA P (P45984) D (44138048)				

Fig. 7. The Source-Domain Sample Visualizations (SDSVs) and Target-Domain Sample Visualizations (TDSVs), which display the high-response regions of CPInformer and MMDG-DTI for the samples in the test set. CPInformer highlights the high-response regions of proteins and drugs in green and blue. MMDG-DTI highlights the corresponding regions in yellow and orange. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The experimental results verified the superiority of our method over the state-of-the-art approaches in both single-domain and cross-domain tasks. While the proposed MMDG-DTI framework demonstrated improved generalization capabilities, there is still room for improvement, particularly regarding predictive accuracy across various cross-domain experiments. In the future, we will further optimize the proposed model based on cutting-edge methods, such as larger language models and more advanced domain generalization methods, providing the field with more reliable and versatile DTI prediction models to empower AI for drug discovery.

CRedit authorship contribution statement

Yang Hua: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhenhua Feng:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis. **Xiaoning Song:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Xiao-Jun Wu:** Supervision, Project administration, Funding acquisition, Writing – review & editing. **Josef Kittler:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets and codes will be available on the project homepage: <https://github.com/JU-HuaY/MMDG-DTI>.

Further details on data availability are presented below.

Dependency

The code has been tested in the following environment:

Python-3.8,

PyTorch-1.8.0,

CUDA-11.1,

Dataset link: https://drive.google.com/file/d/1TLcGFcpmgFIPqm1aBw-EW23TdOy0z7rL/view?usp=share_link.

BindingDB_Cold dataset: https://drive.google.com/file/d/1hSsu2D8XcRama_gnmp5FpxSEXi_wtbOV/view?usp=drive_link.

Trained models link: <https://drive.google.com/file/d/1dSpGwBbFp6yOPTClY4a-LbjvWJrowhdY/view?usp=sharing>.

Store the train model in a path “PATH_MODEL”, the model could be loaded by the following code in main.py

```
"model_data = torch.load(PATH_MODEL)"
```

Data_process:

The original data is first processed by the code document "Pre-data.py"; Then we can use "data_split_name.py" to split the data sets; After that, we can run main.py. This file will also import "merge.py" and "model.py"; The only thing we have to do is to write the data protocols and the save path; The specific data protocols are described in the file "data_merge.py";.

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